Chapter - 2
Theory of Learning

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1 Can We Learn?

Revisiting the process of learning introduced in chapter one : We have the final hypothesis function h_f chosen by the learning algorithm using the data points available to us. However, an important question arises : How can we be sure that a hypothesis function that performs well on the available data points, will perform well out-of-sample, on fresh unseen data points?

2 Probability to the Rescue

In order to tackle the fundamental question raised above, let us continue with the loan example from chapter one and frame it as a simple probability experiment. Think of the general population of a city as a pot of marbles, where each marble represents an individual customer, as illustrated in the image below :

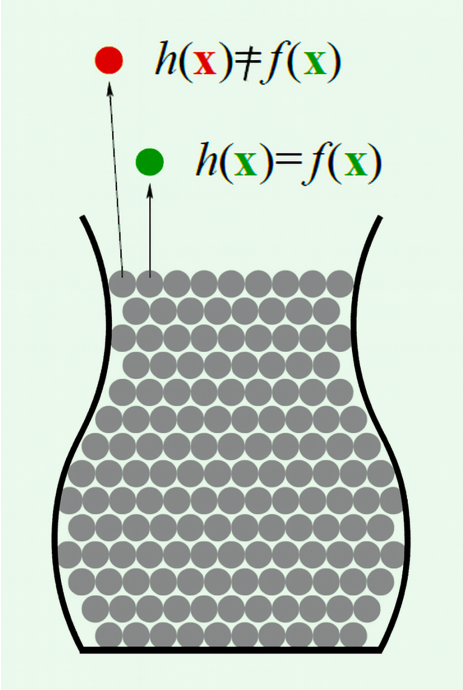


Figure 1: The Marble Pot

Grey marbles represent fresh unseen data points, Green marbles represent correct predictions by our hypothesis function and the Red marbles represent errors. Finally, our data-set corresponds to a sample drawn from that pot.

In line with the above example, $E_{in}(h)$ the in-sample error of our hypothesis function, computed as per the learning model employed, corresponds to the fraction of red marbles in our sample. Whereas, the unknown out-of-sample error $E_{out}(h)$ will correspond to the unknown fraction of red marbles inside the pot, as shown below :

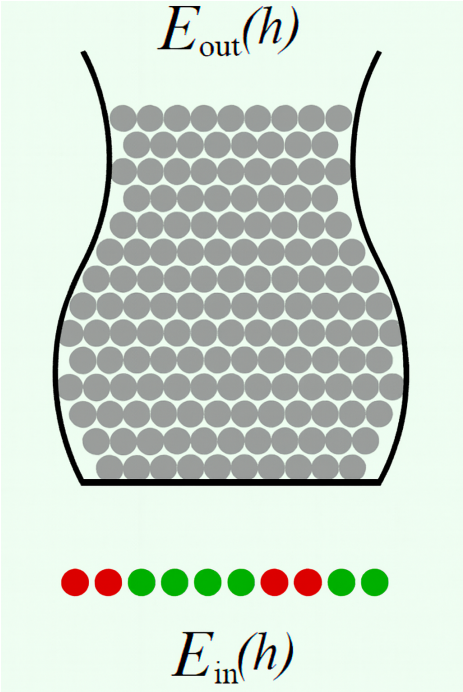


Figure 2: Possible Vs Probable

Now, let us consider a specific bin consisting of 100 marbles, 90 green and 10 red. Suppose we pick a sample of 10 marbles out of that bin independently. Is it possible that all of them or most of them turn out to be red?

Well, it is **possible**, but is it **probable**? Not really!

Computing the exact probability shows that the chances of getting 5 or more reds in a sample of 10 from that bin, is only about 0.16%. In other words, picking a mostly red sample from a mostly green bin is highly unlikely.

How the above probability example connects to the learning scenario ? The above example shows that, as per the laws of probability, we can expect $E_{in}(h)$ and $E_{out}(h)$ to track each other, with a large enough sample-size.

This probabilistic understanding is formalized by the **Hoeffding Inequality** as follows :

$$P[\ |E_{in}(h) - E_{out}(h)| > \epsilon\ ] \leq 2e^{-2\epsilon^2 N}$$

In simple words :

The probability of E_{in} and E_{out} differing from each other will be low, if we accept a reasonable approximation and have a lot of data points.

Let us clarify the terms in the above inequality :

- $E_{in}(h)$: In-sample error of a hypothesis.
- $E_{out}(h)$: Unknown out-of-sample performance of a hypothesis.
- ϵ : The tolerance value, which is the maximum acceptable difference between E_{in} and E_{out}.
- N : Number of available data points.

It can be observed from the above expression that a **large** N makes the probabilistic bound tighter, giving us more confidence for out-of-sample performance, whereas a **small** ϵ makes the bound looser, since such precision is harder to guarantee.

3 Connection to Learning

In the previous section, we have established that E_{out} is likely to track E_{in} for a given hypothesis h. However, in a real learning scenario, we are not just given a final hypothesis function directly. We need to search through the hypothesis set, based on the data points in order to find the best hypothesis function. The real learning scenario is depicted in the following image :

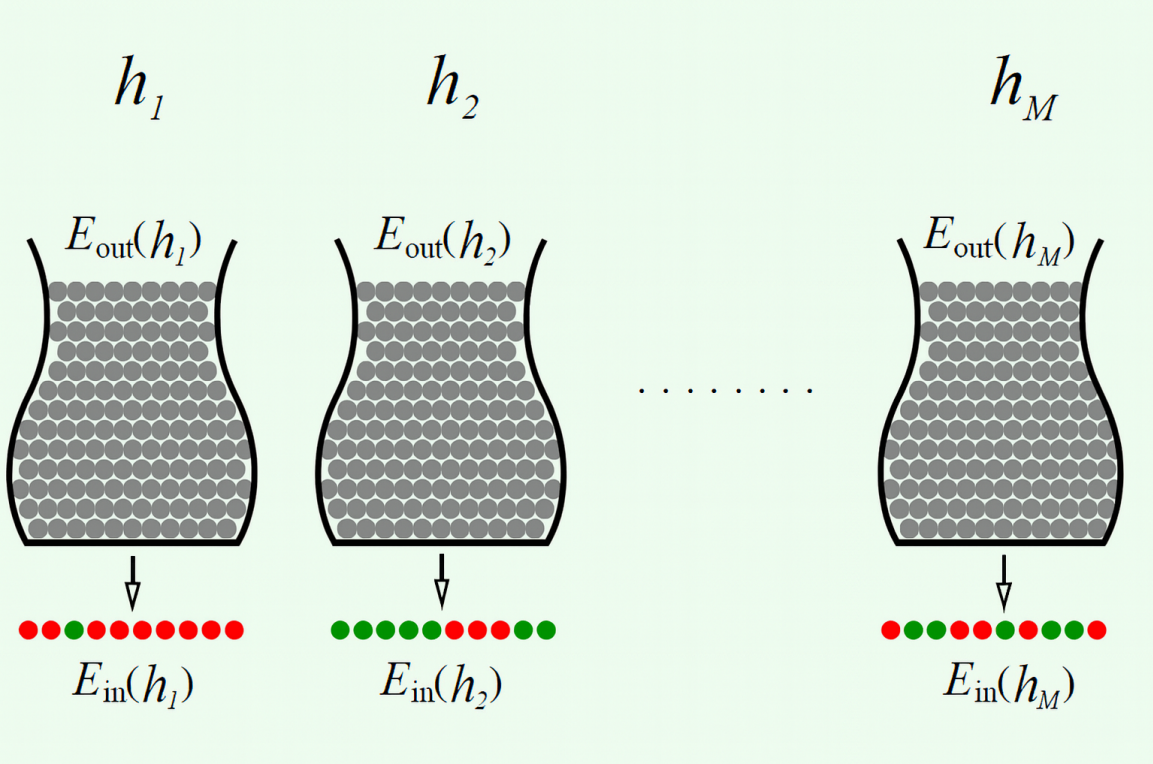


Figure 3: Multiple Hypothesis

Each hypothesis defines it own pot as the fraction of red marbles depends on h, both Ein(h) and Eout(h). So does Hoeffding inequality apply to the multiple pot

scenario as well. Let us continue with our probability example. If we consider 20 bins, each mostly green. What is the probability of finding a sample that has 5 or more reds. We see that the chances have increased markedly. This is because when considering multiple hypothesis, the chances of one of them going wayward by fluke increases as we increase the number of hypothesis under consideration. Hence as per the laws of probability, we will need to account for that in the Hoeffding inequality. Finding the exact probability in the learning situation will require us to look closely at hypothesis set and the data points to figure how different hypothesis functions overlap with each other. For now, we look at a simpler solution, we can ignore the overlaps between different hypotheses and use the union bound to bound the overall probability of the learning process.

This adds an M term. Which might be infinite for simple models like perceptron. We will see that there is a way to bound probability even when possible hypothesis are infinitely many. For that we will have to consider the overlap between the different hypothesis functions and their performance as visible to us on the available data points.

Plan

How to structure the chapter going forward.

Lecture 5 : Hypotheses can be infinitely many. But we need to first look at which are the relevant hypothesis that need to be counted = Dichotomies, Growth Function and Break Point.

Lecture 6 : Through which we will arrive at the the VC inequality. Which is a modification to the Hoeffding Inequality that captures the real learning scenario.

Lecture 7 : Finally, we finish with the final theory notations of VC dimension and Generalization Bound. Lecture 8 is filler as well just like lecture 4, might just add a few points from that.

4 References

1. CalTech Machine Learning Course - CS156, Lecture 2.
2. CalTech Machine Learning Course - CS156, Lecture 4.
3. CalTech Machine Learning Course - CS156, Lecture 5.